Investment Management — Complete Exam Guide

How to Understand Every Topic and How to Answer Every Question
PGDM Term IV Mid-term (Open Book) · Prof. Anil Misra

Prepared for your mid-term

Contents

How to use this guide **3**

PART 1 — How to actually win this exam **4**
1.1 The grading scheme, decoded . . . 4
1.2 The universal answer template . . . 4
1.3 Time and paper strategy . . . 5
1.4 What "illustration" means (the 40% component) . . . 5

PART 2 — Question 1 type: Risk, Return & Choosing a Stock **6**
2.1 Returns — and why there are so many kinds . . . 6
2.2 Risk — measuring uncertainty . . . 7
2.3 Judging return against risk (the heart of Q1) . . . 7
2.4 Value at Risk (VaR) — "the worst case I should prepare for" . . . 8
2.5 Is the data "sufficient"? (part a) . . . 8
2.6 Passive vs active investing (part c) . . . 9
2.7 Fully worked Question 1 . . . 9

PART 3 — Question 2 type: Duration & Convexity **10**
3.1 First, the bond and its price . . . 10
3.2 Duration — what it actually is . . . 10
3.3 Modified Duration — turning duration into a price move . . . 11
3.4 The direction-of-change question . . . 11
3.5 Convexity — the correction for big moves . . . 12

PART 4 — Question 3 type: Immunisation **13**
4.1 The problem: two risks that can cancel . . . 13
4.2 The two conditions for immunisation . . . 13
4.3 The four steps . . . 13
4.4 Fully worked Question 3 . . . 14

PART 5 — Question 4 type: Concept notes (where explanation is everything) **15**
5.1 Market efficiency — the big idea . . . 15
5.2 "Even efficient markets have pockets of inefficiency" — do you agree? (Q4a) . . . 15
5.3 Clean price vs dirty price (Q4b) . . . 16
5.4 Event study methodology (Q4c) — the standard test of semi-strong efficiency 16

5.5 Direct tests of weak-form efficiency (could be asked, and appears in your Excel) 17

PART 6 — Supporting topics that can appear **18**
6.1 Term structure of interest rates (the yield curve) 18
6.2 Bond types and the risks they carry 18
6.3 Market microstructure — how trades actually happen 18
6.4 Behavioural finance — why the inefficiencies exist 19
6.5 Technical vs fundamental analysis (ties back to efficiency) 19

PART 7 — Complete model answers to the 2023-24 paper **20**
Question 1 — Britannia vs Dabur vs NIFTY 20
Question 2 — Shree Transport bonds (Coupon 12%, YTM 12%, Maturity 15 yr, Macaulay D 10 yr, semi-annual) 21
Question 3 — Immunisation (Rs 100 mn in 7 years, rate 12%) 21
Question 4 — Concept notes 21

PART 8 — Formula appendix and night-before checklist **23**
Every formula in one place 23
The decision criterion for each question type 23
Night-before checklist 23

How to use this guide

This is not a formula sheet. It is a teaching guide built from your slides (PPT 1 and 2), the Reilly/Damodaran readings, the Excel models your professor used in class, and — most importantly — the **solved 2023-24 paper**, which tells us exactly what he asks and how he wants it answered.

Each topic is explained in four layers, in this order:

1. **The idea in plain English** — what the concept really means and why it exists. If you understand this, you can answer even a question you have never seen.
2. **The formula, with every symbol explained** — so a formula is never a black box.
3. **A fully worked example** — every step spelled out, so you can copy the *method*, not just the answer.
4. **The model answer** — written the way you should write it in the exam, and a note on the mistakes that lose marks.

Read Part 1 first (it is short and it is worth easy marks on every question). Then work through the part matching each of the four questions. Part 7 gives you the complete, exam-ready answers to the 2023-24 paper — read these last, once the concepts make sense.

PART 1 — How to actually win this exam

1.1 The grading scheme, decoded

Your professor grades every question in **steps**, and he has told you the weights. This is the single most important thing in this guide, so read it twice:

Component	Weight	What it means for you
Correct interpretation of the situation	30%	Show you understood *what is being asked* and *what the numbers mean*.
Approach to the problem	20%	State the method/logic you will use *before* you compute.
Suitable model / illustration / explanation	40%	The formula, the labelled working, a small table or diagram.
Final answer (the number)	10%	The actual result.

The number is worth only 10%. Ninety percent of the marks are for *thinking visibly*. This completely changes how you should write.

- If you set the problem up correctly, write the formula, and explain your logic — **but make an arithmetic slip** — you still earn up to 90%.
- If you write only the final number with no working, you cap yourself at ~10% *even if the number is perfect*.

So the golden rule is: **never write a bare answer.** Always show the four moves — interpret, approach, model, answer.

1.2 The universal answer template

Use this skeleton for *every* question, numerical or conceptual. It maps one-to-one onto the marking scheme:

> **(1) Interpret** — one line restating what is being asked and what matters. *"We must compare two stocks on a risk-adjusted basis, not just on return."*
>
> **(2) Approach** — one line naming the method. *"I will compute mean return, standard deviation, and the return-per-unit-risk ratio for each, then choose the higher ratio."*
>
> **(3) Model / working** — the formula, then a small labelled table with the numbers plugged in.
>
> **(4) Interpret the result & conclude** — say what the number means and give the verdict. *"Britannia earns 0.43 units of return per unit of risk versus Dabur's −0.03, so Britannia is preferred."*

The professor literally rewards steps (1) and (2) with 50% of the marks, and most students skip them. Do not skip them. They take two sentences.

1.3 Time and paper strategy

- **2 hours, 4 questions, 20 marks.** That is roughly **30 minutes per question**, or about 1.5 minutes per mark. Do not overrun on Q1's arithmetic.
- **Do the question you know best first** to bank marks and settle your nerves. Q4 (concepts) is often the fastest for easy marks — consider starting there.
- **"Brevity shall be rewarded"** is printed on the paper. Write in tight, labelled steps and short paragraphs. A page of waffle scores *less* than half a page of clear structure.
- It is **open book.** Bring: this guide, the two Excel models (PPT 1 and PPT 2), and the solved paper. Tab the immunisation and duration pages — those are the ones you will flip to under time pressure.

1.4 What "illustration" means (the 40% component)

Whenever you can, draw or tabulate. A tiny sketch of a yield curve, a duration table with four rows, a price-yield curve showing convexity, a timeline of cash flows — each of these is worth real marks because it is the "suitable illustration." You do not need art; a labelled box-and-arrow diagram counts.

PART 2 — Question 1 type: Risk, Return & Choosing a Stock

This is the data question: you are handed a table of monthly prices for two stocks and an index, and asked to analyse them. It is worth the most marks (8 of 20), so master it. Everything here comes straight from the "Auto/Finance/IT/FMCG sector" Excel model your professor built.

2.1 Returns — and why there are so many kinds

A **return** is simply the percentage gain from holding an asset. But the exam expects you to know *which* return to use and *why*.

Period return (the building block):

$$R_t = \frac{P_t - P_{t-1} + D_t}{P_{t-1}}$$

where P_t is this period's price, P_{t-1} last period's, and D_t any dividend. When only prices are given (as in the exam table), you compute the **capital-gain part only** and must *note that dividends are ignored*, so the true total return is understated. That single sentence earns "interpretation" marks.

Expected (forward-looking) return = Arithmetic Mean (AM):

$$\bar{R} = \frac{1}{n}\sum R_t$$

We use the AM as the *expected* return for next period because each period is treated as an independent draw; the AM is the best estimate of a single future period.

Realised (backward-looking) return = Geometric Mean (GM):

$$GM = \left(\frac{P_n}{P_0}\right)^{1/n} - 1$$

The GM is the *actual compounded* growth you earned over the whole window. It is always $\leq$ AM, and the gap grows with volatility. Rule of thumb from your notes: **AM for expected return, GM for historical/holding-period return.**

Gross vs Net: Net return = Gross return − transaction costs − information costs. Also distinguish **pre-tax vs post-tax** and **nominal vs real**.

Real (inflation-adjusted) return — the Fisher equation:

$$1 + R_{real} = \frac{1 + R_{nominal}}{1 + i} \quad \Rightarrow \quad R_{real} \approx R_{nominal} - i$$

Example: a bond yielding 12% when inflation is 6% gives a real return of $1.12/1.06 - 1 = 5.66\%$, not 6%. This matters most for debt and index-linked bonds.

2.2 Risk — measuring uncertainty

Risk is the *spread* of returns around the mean. We measure it with the **standard deviation** (the square root of variance):

$$\sigma = \sqrt{\frac{\sum (R_t - \bar{R})^2}{n-1}}$$

We divide by $n-1$ (not n) because we are estimating from a *sample* — this is the `STDEV.S` function in Excel. A higher σ means returns bounce around more, so the outcome is less certain. Standard deviation is in the same units as the return (percent), which is why we can compare it directly to the mean.

Intuition: two stocks can have the same average return, but the one with the larger σ can hand you a much worse month. Risk is what you are *compensated* for bearing — hence the next section.

2.3 Judging return against risk (the heart of Q1)

More return is good; more risk is bad. To choose between stocks you must combine the two into a **risk-adjusted** measure. Your professor's framework uses these:

Return-to-Risk ratio (R-T-R) — the key decision criterion in your course:

$$\text{R-T-R} = \frac{\bar{R}}{\sigma}$$

It answers: *how many units of return do I get for each unit of risk I take?* Higher is better. In the class Excel it is literally labelled "Most Imp." This is the number you should lead with when choosing a stock.

Coefficient of Variation (CV) — the same idea flipped over:

$$CV = \frac{\sigma}{\bar{R}} = \frac{1}{\text{R-T-R}}$$

It is *risk per unit of return*, so here **lower is better**. Note: if the mean return is negative, CV is meaningless (say so — it is an interpretation mark).

Sharpe ratio — R-T-R with the risk-free rate removed, the "textbook" version:

$$\text{Sharpe} = \frac{\bar{R} - R_f}{\sigma}$$

It measures *excess* return (above a safe asset) per unit of risk. Use it if a risk-free rate is given; otherwise R-T-R is the course default.

Beta (β) — sensitivity to the market:

$$\beta = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)} = \text{SLOPE}(\text{stock returns}, \text{market returns})$$

Beta tells you how much the stock moves when the market moves 1%. $\beta = 1$ moves with the market; $\beta > 1$ is aggressive (amplifies market swings); $\beta < 1$ is defensive. A low beta with a high return is the ideal — the stock made money without taking much *market* risk.

Correlation (ρ) — how tightly two series move together:

$$\rho = \frac{\text{Cov}(R_i, R_m)}{\sigma_i \, \sigma_m}, \qquad -1 \le \rho \le 1$$

Low correlation between assets is what makes diversification work.

2.4 Value at Risk (VaR) — "the worst case I should prepare for"

This is exactly how the professor phrases part (d): *"the maximum loss you should be prepared for."* VaR turns risk into a rupee figure.

The logic: assume monthly returns are roughly **normally distributed**. On a normal curve, 95% of outcomes lie above the point mean $- 1.645\sigma$ (because 1.645 is the z-value that cuts off the bottom 5% — in Excel, `NORMSINV(0.95) = 1.645`). So the *worst return you would expect in 19 months out of 20* is:

$$\text{VaR}\% = -\bar{R} + 1.645\,\sigma$$

(The minus sign on the mean is because a positive average return *cushions* the loss.) Then:

$$\text{VaR in Rs} = \text{VaR}\% \times \text{amount invested}$$

How to present it fully (earns all the marks):

- State the assumption: returns $\sim$ Normal, 95% confidence.
- Compute the 95% VaR and convert to rupees.
- Add the **more extreme** bounds: a 3-sigma month ($\bar{R} - 3\sigma$, ~99.7% confidence) and the **absolute maximum** — in equity you can lose the *entire* investment.

At 99% confidence use $z = 2.33$ instead of 1.645.

2.5 Is the data "sufficient"? (part a)

The professor asks you to judge the data on three named dimensions. Learn these three words — **Nature, Length, Frequency** — and you can answer this every time:

- **Nature:** *What kind of data is it?* Only closing prices → captures capital gains but **ignores dividends** (total return understated); prices are **unadjusted** for splits/bonuses; there is **no fundamental data** (earnings, debt) and **no risk-free rate** to compute Sharpe. So the raw material is thin.
- **Length:** *How long is the window?* 13 months = 12 return observations = **one market phase**, not a full market cycle. Contrast the class study, which used ~3 years (≈155 observations). A short window gives unstable, unreliable estimates.
- **Frequency:** *How often is it sampled?* Monthly is acceptable in principle, but 12 points is a **tiny sample** — the standard error of the mean, SD, beta and VaR is large, so you cannot trust them for a real decision.

Conclusion to write: the data allows only a *first approximation*; a sound decision needs a longer, higher-frequency, dividend-adjusted series plus fundamentals.

2.6 Passive vs active investing (part c)

Passive = buy and hold the index (here, NIFTY 50); accept the market return, pay almost nothing, and get full diversification. **Active** = pick individual stocks to try to beat the index.

How to argue it *with this data*: compare the chosen stock's R-T-R to the index's. If the stock's risk-adjusted return beats the index, active *won* over this window — but always add the caveats: (i) the index has the lowest SD and VaR because of **diversification**; (ii) active only wins **if you pick the right stock** — the wrong pick (Dabur) would have underperformed the index; (iii) passive protects an unskilled investor cheaply. This balanced answer (with the efficient-markets link — see Part 5) is what scores.

2.7 Fully worked Question 1

Data given: 13 monthly closing prices for Britannia, Dabur and NIFTY 50 (Apr-22 to Apr-23). Assume the future resembles 2022-23.

Step 1 — turn 13 prices into 12 returns using $R_t = (P_t - P_{t-1})/P_{t-1}$. For Britannia, May-22 $= (3648 - 3279)/3279 = +11.25\%$; Jun-22 $= (3466 - 3648)/3648 = -4.99\%$; and so on.

Step 2 — compute the summary statistics for each column (mean, SD, then the ratios, beta and VaR):

Measure (monthly)	Britannia	Dabur	NIFTY 50
Average return (AM)	2.98%	−0.18%	0.55%
Risk (SD, sample)	6.93%	6.61%	4.44%
R-T-R = Mean / SD	**0.431**	−0.027	0.123
Coefficient of Variation	2.32	negative (n/m)	8.13
Beta vs NIFTY	0.60	1.07	1.00
VaR 95% = −Mean + 1.645·SD	8.41%	11.05%	6.75%
VaR on Rs 1,00,000	Rs 8,412	Rs 11,053	Rs 6,752

(VaR check, Britannia: $-2.98 + 1.645 \times 6.93 = 8.41\%$*.)*

The four parts are then answered from this one table — see the complete model answers in **Part 7, Question 1**.

Common mistakes: (i) using n instead of $n - 1$ for SD; (ii) forgetting the minus sign on the mean in VaR; (iii) choosing the stock with the higher *return* instead of the higher *R-T-R*; (iv) giving a one-word "yes/no" on data sufficiency instead of the Nature/Length/Frequency structure.

PART 3 — Question 2 type: Duration & Convexity

Q2 gives you a bond's characteristics and asks about **interest-rate risk** — how much the price moves when yields move. To answer it you must truly understand duration, not just quote $D/(1+y)$.

3.1 First, the bond and its price

A bond is a loan: the issuer pays you periodic **coupons** and returns the **face value** at maturity. Its fair price is the present value of those cash flows:

$$P = \sum_{t=1}^{n} \frac{C}{(1+y)^t} + \frac{F}{(1+y)^n}$$

C = coupon in rupees, F = face value, y = yield to maturity (the market discount rate), n = number of periods. For **semi-annual** bonds, halve the coupon and yield and double the periods.

Why price and yield move in opposite directions: y sits in the denominator. If market yields rise, every cash flow is discounted harder, so the present value — the price — **falls**. This inverse relationship is the source of all interest-rate risk.

Three facts you will use (all provable from the formula):

- **Longer maturity → more price-sensitive** (cash flows are discounted over more periods).
- **Lower coupon → more price-sensitive** (more of your money is locked in the distant final payment).
- Price rises **more** when yields fall than it drops when yields rise by the same amount (**convexity/asymmetry**).

3.2 Duration — what it actually is

Two bonds can have the same maturity but very different risk, so *maturity is a poor risk measure*. **Duration** fixes this. Think of it as the **balance point (centre of mass) of the bond's cash flows on a time line** — the weighted-average time, in years, until you get your money back, where each time is weighted by how much of the present value arrives then.

$$D = \sum_t \left[\frac{PV(C_t)}{P} \times t \right] = \sum_t (w_t \times t), \qquad w_t = \frac{PV(C_t)}{P}$$

How to compute it by hand (the table method — learn this):

Build five columns. Example: a **3-year, 8% annual-coupon, Rs 1,000 bond, YTM 10%.**

t (yr)	Cash flow C_t	$PV = C_t/1.10^t$	Weight $w_t = PV/P$	$w_t \times t$
1	80	72.73	0.0765	0.0765
2	80	66.12	0.0696	0.1392
3	1080	811.42	0.8539	2.5617

t (yr)	Cash flow C_t	$PV = C_t/1.10^t$	Weight $w_t = PV/P$	$w_t \times t$
		P = 950.26	**1.000**	**D = 2.777**

So the duration is **2.777 years** — less than the 3-year maturity, because the coupons pull the balance point earlier. (For a *zero*-coupon bond there are no early cash flows, so its duration equals its maturity exactly.)

3.3 Modified Duration — turning duration into a price move

Duration is measured in years; **modified duration (D^* or MD)** converts it into a **percentage price change per 1% change in yield**:

$$MD = \frac{D}{1 + y/n}$$

where n = coupon payments per year. Then the price change for a small yield change Δy is:

$$\%\Delta P \approx -MD \times \Delta y$$

The minus sign is the inverse relationship. So if $MD = 9.43$, a $+1\%$ yield shock cuts the price by about 9.43%.

Worked (the actual Q2 numbers): Macaulay duration $= 10$ years, YTM $= 12\%$ paid semi-annually ($n = 2$):

$$MD = \frac{10}{1 + 0.12/2} = \frac{10}{1.06} = 9.43 \text{ years}$$

Why modified duration beats maturity (write this): maturity looks only at the timing of the *final* payment and ignores the size and timing of every coupon in between. Duration weights *all* cash flows by their present value, and modified duration converts that into an actual price sensitivity. Two bonds with the same maturity but different coupons have different risk — maturity cannot see that difference; modified duration can.

3.4 The direction-of-change question

The professor often asks: *how does modified duration change if…* Reason it from the two properties:

- **Coupon 6% instead of 12% → duration INCREASES.** Smaller coupons mean less money arrives early, so proportionately more weight sits on the distant final payment → the balance point moves later → duration (and MD) rises. *Coupon and duration move inversely.*
- **Maturity 10 years instead of 15 → duration DECREASES.** All cash flows arrive sooner → earlier balance point → shorter duration. *Maturity and duration move together.*

3.5 Convexity — the correction for big moves

Modified duration draws a **straight line** through the price-yield curve at today's yield. But the true relationship is a **curve that bends** (it is *convex* to the origin). So the straight-line estimate is only accurate for *small* yield changes. For large ones it **overstates the price fall** and **understates the price rise**.

Convexity measures that bending (it is the second derivative of price with respect to yield, scaled by price — the rate at which duration itself changes as yields move). We add a second-order term:

$$\%\Delta P \approx \underbrace{(-MD \times \Delta y)}_{\text{duration effect}} + \underbrace{\left(\tfrac{1}{2} \times \text{Convexity} \times (\Delta y)^2\right)}_{\text{convexity correction}}$$

The convexity term is **always positive** for a normal bond, which is *good news* for the holder: it means your bond falls a little less than duration predicts when yields rise, and gains a little more when they fall.

Worked: $MD = 9.43$, convexity $= 120$, yields rise $\Delta y = +1\%$:

$$\%\Delta P \approx -9.43\% + \tfrac{1}{2}(120)(0.01)^2 = -9.43\% + 0.60\% = -8.83\%$$

So duration alone predicted a 9.43% fall; convexity softens it to 8.83%.

Common mistakes: forgetting to divide by $(1 + y/n)$ with the *semi-annual* adjustment; dropping the minus sign; and, on convexity, forgetting the $\frac{1}{2}$ and that $(\Delta y)^2$ makes the term positive whether yields rise or fall.

PART 4 — Question 3 type: Immunisation

Q3 is the "insurance company / pension fund" problem: you owe a fixed sum on a future date and must build a bond portfolio that is **protected against interest-rate changes**. This is the highest-value conceptual-plus-numerical question, and students fear it — but it is just duration matching.

4.1 The problem: two risks that can cancel

If you must pay a fixed amount in, say, 7 years and you buy bonds to fund it, you face **two opposing risks** when interest rates change:

- **Price risk:** if rates *rise*, your bonds are worth less when you might need to sell them. (Bad when rates rise.)
- **Reinvestment risk:** if rates *fall*, the coupons you receive get reinvested at lower rates, so you accumulate less. (Bad when rates fall.)

These two risks pull in **opposite directions**. **Immunisation** is the trick of balancing them so they exactly cancel, leaving your final wealth unchanged whichever way rates move — like a vaccine that leaves the portfolio immune to rate shocks.

4.2 The two conditions for immunisation

A portfolio is immunised against a liability when:

1. **Present values match:** $PV(\text{assets}) = PV(\text{liability})$ — you have funded the right amount today; **and**
2. **Durations match:** $D(\text{assets}) = D(\text{liability})$ — the assets and the liability have the same interest-rate sensitivity, so they move together.

A single future payment (like the liability) has a duration equal to its **time to payment**. So to immunise a 7-year obligation, build an asset portfolio with duration = 7 years.

You will need the durations of the two instruments you are given:

- **Zero-coupon bond:** duration = its maturity.
- **Perpetuity** (pays a coupon forever): $D_{perp} = \dfrac{1+y}{y}$.

4.3 The four steps

1. **Find the PV of the obligation:** $PV = \dfrac{FV}{(1+y)^n}$.
2. **Solve for the weights** that make the portfolio duration equal the target: $w \cdot D_1 + (1-w) \cdot D_2 = D_{target}$, with the weights summing to 1.
3. **Convert weights to money:** amount in each instrument \$= \$ weight $\times PV$(obligation).
4. **Rebalance over time:** as time passes and rates change, the durations drift out of line, so you must periodically readjust. **Immunisation is a dynamic strategy, not "set and forget."**

Feasibility check (an easy interpretation mark): the weighted average of the two durations can only equal the target if **one instrument's duration is below the target and**

the other's is above. If both are on the same side, a weight comes out negative — which would require short-selling.

4.4 Fully worked Question 3

Given: obligation of **Rs 100 million in 7 years**; market rate **12%**; fund it with **zero-coupon bonds and perpetuities** paying annual coupons.

Part (a) — PV of the obligation:

$$PV = \frac{100}{(1.12)^7} = \frac{100}{2.2107} = \text{Rs } 45.23 \text{ million}$$

Part (b) — how to immunise, and the amounts. Target duration = 7 years. Perpetuity duration $= 1.12/0.12 = 9.33$ years.

Important subtlety the professor flags: the paper says **10-year** zeros. A 10-year zero has duration 10, and the perpetuity has duration 9.33 — **both are above the 7-year target**, so the equation $w(10)+(1-w)(9.33) = 7$ gives $w = -3.5$, a **negative weight**: you cannot immunise with two long-duration assets held long; you would have to *short* the zeros. The standard, feasible version of this problem (and the one you should present) uses a **3-year zero** ($D = 3$, below the target):

$$w(3) + (1-w)(9.33) = 7 \;\Rightarrow\; 9.33 - 6.33w = 7 \;\Rightarrow\; w = 0.368$$

So invest:

- **Zeros:** \$0.368 ×45.23 = \$ **Rs 16.66 million**
- **Perpetuity:** \$0.632 ×45.23 = \$ **Rs 28.57 million**

Show the feasibility remark either way — it earns interpretation marks and shows you understand duration matching rather than just plugging numbers.

Part (c) — one year later, rates fall to 11%; still fully funded? The obligation is now 6 years away, so it has grown in present value:

$$PV_{liability} = \frac{100}{(1.11)^6} = \text{Rs } 53.46 \text{ million}$$

Revalue the assets at 11%: the zeros (now 2-year instruments) plus the perpetuity plus the coupon just received come to about **Rs 53.60 million**. Since assets (53.60) $\geq$ liability (53.46), the position is **still fully funded**, with a small surplus of ~Rs 0.14 million arising because the asset portfolio has *more convexity* than the single-payment liability. **But** the durations have drifted (liability 6, zero 2, perpetuity $1.11/0.11 = 10.09$), so you must **re-balance**: solve $w(2)+(1-w)(10.09) = 6 \Rightarrow w \approx 0.506$. This demonstrates the key lesson: immunisation must be **re-checked and rebalanced** as conditions change.

The three levels of matching (mention as a closing line): match the **amount** (PV), match the **duration**, and — to guard against large or non-parallel yield-curve shifts — match the **convexity** too.

PART 5 — Question 4 type: Concept notes (where explanation is everything)

Q4 asks for short notes "with suitable illustrations." Here the *explanation* is the whole answer, so this is where a formula sheet fails you and understanding pays off. The three that appeared last year — pockets of inefficiency, clean vs dirty price, and event studies — are all likely again.

5.1 Market efficiency — the big idea

An **efficient market** is one in which the price of a security is an **unbiased estimate of its true value** because it already reflects all available information. New information is absorbed into prices **almost instantly**, so at any moment the price is "fair." A consequence is that price changes are essentially **random** (a "random walk") — because only *new* (unpredictable) information moves prices, and if it were predictable it would already be in the price.

Fama's three forms differ by *what information* is already in the price:

Form	Information reflected in price	Which analysis becomes useless
Weak	All past prices and volumes	**Technical analysis** (charts, patterns) cannot beat the market
Semi-strong	All *public* information (accounts, news, announcements)	**Fundamental analysis** cannot beat the market either
Strong	*All* information, including private/insider	Even insiders cannot earn excess returns

Each form contains the previous one. Most evidence says developed markets are **weak-form and largely semi-strong efficient**, but **not strong-form** (insiders *do* profit).

What efficiency implies: it is very hard to beat the market consistently, so it favours **passive/index investing**, minimal trading, and broad diversification. **What it does NOT imply:** that prices never change, that no one ever profits, or that the price equals true value at every instant with zero error — it means there is **no *systematic, exploitable* mispricing** after costs.

5.2 "Even efficient markets have pockets of inefficiency" — do you agree? (Q4a)

Yes — and here is the reasoning to write:

Efficiency is a **continuum**, not an on/off switch. Three arguments:

1. **The Grossman-Stiglitz paradox (the deepest point).** If prices already reflected *all* information perfectly, then no one would have any incentive to spend time and money gathering and analysing information — because they could not profit from it. But if nobody does that analysis, prices would *stop* being efficient. So a *little* mispricing **must**

persist, precisely to reward the analysts and arbitrageurs who keep prices efficient. A price gap smaller than the cost of trading on it is not a true inefficiency — it is the fee the market pays for efficiency.

2. **Efficiency varies by segment.** Large, heavily-covered stocks are very efficient; but **small-caps, neglected or thinly-traded stocks, and emerging markets** — where analyst coverage, disclosure and the ability to short/arbitrage are weaker — stay relatively inefficient.
3. **Documented anomalies and behavioural biases.** The January/size effect, momentum, and post-earnings-announcement drift, plus biases like herding and over-/under-reaction, create *temporary* mispricings.

Illustration (use this): the **Numeric Investors** case — a quant firm earned excess returns by exploiting analysts' *underreaction* to earnings news, a genuine pocket of inefficiency in the otherwise highly efficient US market. As competitors copied the strategy, the profit window shrank from weeks to minutes — showing efficiency and inefficiency *co-existing* and the market self-correcting.

5.3 Clean price vs dirty price (Q4b)

The situation: bonds pay coupons on fixed dates, but they trade *every* day. If you buy a bond *between* coupon dates, the seller has earned — but not yet received — the coupon for the days since the last payment. That earned-but-unpaid amount is **accrued interest**, and the buyer must compensate the seller for it.

$$\text{Dirty (full/invoice) price} = \text{Clean (quoted) price} + \text{Accrued Interest}$$

$$\text{Accrued Interest} = \text{Coupon} \times \frac{\text{days since last coupon}}{\text{days in the coupon period}}$$

The buyer actually **pays the dirty price**, but markets **quote the clean price**.

Illustration: a 12% bond with semi-annual coupons pays Rs 60 every six months. Sold 3 months after the last coupon at a quoted (clean) price of Rs 980:

$$\text{Accrued} = 60 \times \tfrac{3}{6} = \text{Rs } 30 \;\Rightarrow\; \text{Dirty price} = 980 + 30 = \text{Rs } 1{,}010$$

Why it matters for decisions: (i) the **dirty price is the true cash outflow**, so it is what you use for yield and return calculations; (ii) the **clean price strips out the mechanical accrual**, so it lets you compare a bond fairly across time and against other bonds. If you compared *dirty* prices you would be misled by a "sawtooth" that just reflects the coupon clock building up and resetting — not any change in value or yield.

5.4 Event study methodology (Q4c) — the standard test of semi-strong efficiency

Why it exists: to test whether the market absorbs **public announcements** quickly and correctly — i.e. to test **semi-strong efficiency**. It measures the *speed and accuracy* of the price reaction to news, which is precisely what "informational efficiency" means.

The method, step by step:

1. **Define the event and event date** ($t = 0$): an earnings release, dividend change, bonus, merger, stock split, etc. Define an **event window** around it (e.g. -10 to $+10$ days).
2. **Estimate the "normal" return** each firm *would* have earned absent the event, using the **market model** fitted over a pre-event *estimation window*:
$$E(R_{j,t}) = \alpha_j + \beta_j R_{m,t}$$
3. **Compute the Abnormal Return** each day: what actually happened minus what was expected:
$$AR_{j,t} = R_{j,t} - (\alpha_j + \beta_j R_{m,t})$$
4. **Aggregate:** average the ARs across all firms for each day (**Average Abnormal Return, AAR**), then **cumulate** them over the window (**Cumulative Abnormal Return, CAR**).
5. **Test significance** (a t-test) to see whether the CAR is reliably different from zero.

How to read the result (this is the part that shows understanding):

- **CAR jumps at $t = 0$ and then stays flat** → the market absorbed the news *instantly and fully* → **semi-strong efficient**; you *cannot* profit by trading *after* the announcement.
- **CAR keeps drifting up (or down) for days/weeks after** → the market absorbed the news *slowly* → **inefficiency**, and a trading rule (buy on the announcement) could earn abnormal profit.
- **CAR moves *before* the event** → information **leaked** / insiders traded early → a strong-form violation.

So the shape of the CAR line around the event *is* the measure of informational efficiency — which is why the event study is the standard tool.

5.5 Direct tests of weak-form efficiency (could be asked, and appears in your Excel)

Two tests from the "Tests of Market Efficiency" workbook:

(a) Serial-correlation (autocorrelation) test. Correlate each day's return with the previous day's (R_t vs R_{t-1}), and with longer lags. If returns were predictable from their own past, the correlation would be large; a correlation **near zero** means past returns give no edge → consistent with a **random walk** → weak-form efficient.

(b) Runs test. Mark each day's price change as $+$ or $-$, then count the **runs** — unbroken streaks of the same sign ($+++$ is one run, then $--$ is the next). Compare the **actual** number of runs to the number **expected** under pure randomness:
$$E(\text{Runs}) = \frac{2\,n_1 n_2}{n_1 + n_2} + 1$$
where n_1 and n_2 are the counts of up-days and down-days. If the actual runs are close to expected (a Z-test formalises "close"), price movements are random → weak-form efficient. Too *few* runs means trends (momentum); too *many* means excessive reversals.

PART 6 — Supporting topics that can appear

6.1 Term structure of interest rates (the yield curve)

The **yield curve** plots yield against time-to-maturity for bonds of the same credit quality. Its shape is explained by three theories — and the exam wants you to interpret a shape using them:

- **Pure Expectations Theory.** Long rates are just the **geometric average of expected future short rates**. So an upward-sloping curve means the market **expects short rates to rise** (often signalling expected growth/inflation); a downward slope means it expects them to fall.

$$(1+y_n)^n = (1+y_{n-1})^{n-1}\,(1+E[\text{short rate}])$$

- **Liquidity Preference Theory.** Expectations **plus a liquidity premium** that grows with maturity, because long bonds carry more price risk and investors demand compensation to hold them. Implication: the curve slopes **up even if rates are expected to stay flat.** So interpret any curve using *both* theories.
- **Segmented Markets / Preferred Habitat.** Supply and demand in each maturity "bucket" set its rate somewhat independently, because different investors (banks vs pension funds) prefer different maturities to match their liabilities.

Forward rate — the future short rate *implied by today's* curve (from data, not opinion):

$$(1+y_n)^n = (1+y_{n-1})^{n-1}\,(1+f_n)$$

Yield curve and the economy: an **inverted** curve (long yields below short) has historically **forecast recessions**, typically 4-5 quarters ahead. The spread (10-year minus 3-month) is used in probit models to predict growth and even bear markets. *Mechanism:* if investors expect a slowdown, they buy long bonds now (pushing long yields down) and sell short — flattening or inverting the curve.

6.2 Bond types and the risks they carry

- **By coupon:** fixed, floating (coupon tied to a benchmark), inverse floater (moves opposite the benchmark), zero-coupon.
- **By redemption feature:** callable (issuer can buy it back — bad for you, so it pays a higher coupon), puttable (you can sell it back — good for you), convertible (swap for shares).
- **International:** Eurobond (issued outside the currency's home country) vs foreign bonds (Yankee = USD in US by a foreigner, Samurai = JPY in Japan, Masala = INR bond sold abroad).
- **Risks:** interest-rate risk (price falls when rates rise), reinvestment risk (coupons reinvested at lower rates), credit/default risk (priced as a spread over the government yield), inflation risk, and call risk. **Rating:** investment grade is BBB/Baa and above; below that is "junk"/speculative.

6.3 Market microstructure — how trades actually happen

- **Bid-ask spread.** The **bid** is the highest price a buyer will pay; the **ask** is the lowest a seller will accept. The **spread** = ask − bid is a cost you pay to trade immediately, and

the **mid-price** = (bid+ask)/2 is the "fair" reference. A **narrow spread = a liquid, actively-traded security.**

- **Impact cost.** A large order cannot all fill at the best price — it "walks the book," filling at progressively worse prices (a volume-weighted average). The **impact cost** = (your average execution price − mid-price) / mid-price measures this slippage. It is a hidden but real transaction cost, and a key reason large investors care about liquidity.
- **Order types.** A **market order** executes immediately at the best available price (speed, no price control); a **limit order** executes only at your price or better (price control, but may not fill); a **stop-loss** becomes a market order once a trigger price is hit, to cap losses.
- **Transaction costs.** Explicit (brokerage, the spread, taxes like STT) plus implicit (impact cost, slippage). They reduce net returns and are central to the passive-vs-active debate: frequent active trading racks up costs that a cheap index fund avoids.
- **Short selling.** You **borrow** shares, **sell** them now, and hope to **buy them back cheaper** later, profiting if the price falls. The danger: your **loss is unlimited** (a price can rise without bound) while your gain is capped at 100% (the price can only fall to zero) — the mirror image of a normal long position, where the most you can lose is your investment.

6.4 Behavioural finance — why the inefficiencies exist

Classical theory assumes rational investors; behavioural finance shows we are **systematically** irrational, and these biases push prices away from value (creating the pockets of inefficiency in 5.2). Know a handful by name and effect:

- **Loss aversion / Prospect theory:** losses hurt about twice as much as equal gains please — so people hold losers too long and sell winners too early (the **disposition effect**).
- **Overconfidence:** investors overrate their own information → overtrading.
- **Anchoring:** clinging to a reference number (e.g. the purchase price).
- **Herding:** following the crowd → bubbles and crashes.
- **Over- and under-reaction:** markets overshoot on some news and are slow on other news → momentum and reversals.

6.5 Technical vs fundamental analysis (ties back to efficiency)

- **Technical analysis** studies past prices and volume — charts, patterns, support/resistance, RSI — assuming patterns repeat. It can only work if markets are **not even weak-form efficient**. Favoured by short-term traders.
- **Fundamental analysis** estimates intrinsic value from earnings, assets and the economy. It can only beat the market if markets are **not semi-strong efficient**. If semi-strong efficiency holds, both approaches fail to earn *excess* returns — which is the case for passive investing.

PART 7 — Complete model answers to the 2023-24 paper

Written the way you should write them: interpret, approach, model, conclude. Keep this style in the exam.

Question 1 — Britannia vs Dabur vs NIFTY

(a) Is the data sufficient for an investment decision?

No, the data is not sufficient; it allows only a first approximation. Judged on the three dimensions: **Nature** — only closing prices are given. These capture the capital-gain yield but **ignore dividends**, so total return is understated; prices are unadjusted for corporate actions; and there are no fundamentals or risk-free rate. **Length** — just 13 months (12 return observations) covering a **single market phase**, not a full cycle; the class framework used ~3 years / 155 observations. **Frequency** — monthly is acceptable, but 12 observations is far too small for stable estimates of mean, SD, beta or VaR (high standard error). **Conclusion:** a sound decision needs a longer, higher-frequency, dividend-adjusted series plus fundamental data.

(b) Which of the two stocks would you prefer, and why?

Britannia. On the class Return-Risk framework: expected return 2.98%/month vs Dabur's −0.18% (Dabur destroyed value, Rs 557 → Rs 533); risk is almost identical (6.93% vs 6.61%). The decisive criterion is **return per unit of risk**: Britannia 2.98/6.93 = **0.431** versus Dabur −0.18/6.61 = **−0.027**. Britannia earns 0.43 units of return for each unit of risk; Dabur earns a *negative* reward for the same risk. It also has the lower worst-case loss (VaR Rs 8,412 vs Rs 11,053). **Decision: invest in Britannia.**

(c) Would passive investing be better?

Passive here means holding the NIFTY index. On R-T-R, Britannia (0.431) beats the index (0.123) by ~3.5×, and its return (2.98% vs 0.55%/month) is higher despite a higher SD — with a beta of only 0.60, it took little market risk to do so. So **on this data, active investing in Britannia was superior to passive.** Caveats: the index has the lowest SD and VaR (diversification), and active only wins *if the right stock is picked* — the wrong pick (Dabur) would have underperformed the index. Passive investing protects an unskilled investor at low cost, but the evidence of 2022-23 favours the active choice.

(d) Maximum loss on Rs 1,00,000 (worst case).

Using Value at Risk, assuming returns are normally distributed. Worst-case monthly return at 95% confidence $= -\bar{R} + 1.645\sigma$. Britannia: $-2.98\% + 1.645(6.93\%) = 8.41\%$ → about **Rs 8,412** in a bad month (95% confidence). A 3-sigma month ($2.98\% - 3 \times 6.93\%$) ≈ −17.8% → ~Rs 17,800; and the theoretical maximum equity loss is the entire Rs 1,00,000. **So be prepared for ~Rs 8,400 in a typical bad month, up to ~Rs 17,800 in an extreme month.**

Question 2 — Shree Transport bonds (Coupon 12%, YTM 12%, Maturity 15 yr, Macaulay D 10 yr, semi-annual)

(a) Modified duration; why better than maturity.

$MD = D / (1 + y/n) = 10 / (1 + 0.12/2) = 10/1.06 = $ **9.43 years.** Interpretation: a 1% rise in yield lowers the price by about 9.43%. It beats maturity because maturity uses only the final payment's timing, whereas duration weights *all* cash flows by present value and modified duration converts that into a price sensitivity — so it correctly distinguishes two same-maturity bonds with different coupons.

(b) Direction of change in modified duration.

(i) Coupon 6% instead of 12% → **MD increases** (lower coupon puts more weight on the distant payment, lengthening duration). (ii) Maturity 10 instead of 15 → **MD decreases** (cash flows arrive earlier).

(c) Convexity and its use.

Convexity measures the curvature of the price-yield relationship — the rate at which duration changes as yields change. Duration alone is a straight-line (first-order) estimate, accurate only for small yield moves; for large moves it overstates the fall and understates the rise. The corrected estimate is $\%\Delta P \approx -MD\,\Delta y + \frac{1}{2}\,\text{Convexity}\,(\Delta y)^2$. E.g. $MD = 9.43$, convexity 120, $\Delta y = +1\%$: $-9.43\% + 0.60\% = -8.83\%$. Positive convexity is desirable — it cushions price falls and boosts price rises.

Question 3 — Immunisation (Rs 100 mn in 7 years, rate 12%)

(a) $PV = 100/1.12^7 = $ **Rs 45.23 million.**

(b) Immunise by matching PV **and** duration (target = 7 years). Perpetuity duration $= 1.12/0.12 = 9.33$. Since a 10-year zero (D=10) and the perpetuity (9.33) are both above 7, that combination needs a negative weight (short position); using the standard **3-year zero**: $w(3) + (1-w)(9.33) = 7 \Rightarrow w = 0.368$. Invest **Rs 16.66 mn in zeros** and **Rs 28.57 mn in the perpetuity.**

(c) After 1 year at 11% (6 years to go): liability PV $= 100/1.11^6 = $ **Rs 53.46 mn**; assets ≈ **Rs 53.60 mn**, so **still fully funded** (small surplus from asset convexity). Durations have drifted, so **rebalance** to $w(2)+(1-w)(10.09) = 6 \Rightarrow w \approx 0.506$. Immunisation is dynamic.

Question 4 — Concept notes

(a) Pockets of inefficiency: *agree* — efficiency is a continuum. Grossman-Stiglitz: some mispricing must persist to reward information gathering; weak segments (small-caps, emerging markets, limits to arbitrage) stay inefficient; anomalies and behavioural biases create temporary gaps. Illustration: Numeric Investors exploiting analyst underreaction until competition shrank the window. *(See 5.2.)*

(b) Clean vs dirty price: Dirty = Clean + Accrued Interest; accrued = coupon × (days since last coupon / days in period). Buyer pays dirty, market quotes clean. Example: 12% bond, Rs 60 semi-annual coupon, 3 months in → accrued Rs 30 → dirty Rs 1,010 on a clean Rs 980. Dirty = true cash cost (for yields); clean = comparable across time. *(See 5.3.)*

(c) Event study: tests semi-strong efficiency by measuring the abnormal return around a public announcement. Define event date, estimate normal return via the market model $R = \alpha + \beta R_m$, compute $AR = $ actual − expected, cumulate to CAR, test significance. Instant jump then flat = efficient; post-event drift = inefficiency; pre-event move = leakage. *(See 5.4.)*

PART 8 — Formula appendix and night-before checklist

Every formula in one place

Returns & risk

- Period return: $R_t = (P_t - P_{t-1} + D)/P_{t-1}$
- Expected = arithmetic mean; Realised/HPR = geometric mean $= (P_n/P_0)^{1/n} - 1$
- Real return (Fisher): $(1+R)/(1+i) - 1 \approx R - i$
- Standard deviation (risk): $\sigma = \sqrt{\sum (R - \bar{R})^2/(n-1)}$
- R-T-R $= \bar{R}/\sigma$ · CV $= \sigma/\bar{R}$ · Sharpe $= (\bar{R} - R_f)/\sigma$
- Beta $= \text{Cov}(i,m)/\text{Var}(m) =$ SLOPE · Correlation $= \text{Cov}/(\sigma_i \sigma_m)$
- VaR% $= -\bar{R} + 1.645\sigma$ (95%); VaR(Rs) = VaR% × amount

Bonds

- Price $= \sum C/(1+y)^t + F/(1+y)^n$; Zero $= F/(1+y)^n$; Perpetuity $= C/y$
- Current yield = annual coupon / price; Dirty = Clean + Accrued
- Macaulay $D = \sum w_t t$, $w_t = PV(C_t)/P$; Zero D = maturity; Perpetuity $D = (1+y)/y$
- Modified duration $MD = D/(1 + y/n)$
- $\%\Delta P \approx -MD\,\Delta y + \frac{1}{2}\,\text{Conv}\,(\Delta y)^2$

Immunisation: $PV(\text{assets}) = PV(\text{liab})$ and $D(\text{assets}) = D(\text{liab})$; solve $wD_1 + (1-w)D_2 = D_{target}$.

Term structure: $(1+y_n)^n = (1+y_{n-1})^{n-1}(1+f_n)$.

Efficiency: $AR = R - (\alpha + \beta R_m)$; $CAR = \sum AR$; Runs $E(R) = 2n_1 n_2/(n_1+n_2) + 1$.

The decision criterion for each question type

- **Choosing a stock (Q1):** highest **R-T-R** (return per unit of risk). Report VaR for "worst case."
- **Interest-rate risk (Q2): modified duration** for the price move; add convexity for large shocks.
- **Funding a liability (Q3): match duration** (and PV); rebalance over time.
- **Is the market efficient (Q4):** look at whether prices adjust **instantly** to news (event study / CAR).

Night-before checklist

1. Can I turn a price column into returns, then compute mean, SD, R-T-R, beta and VaR? (Practise once on the Britannia data.)
2. Can I build a duration table (t, CF, PV, weight, weight×t) from scratch and get $MD = D/(1+y/n)$?
3. Can I do the immunisation four steps — PV, solve weights, rupee amounts, rebalance?
4. Can I explain the three EMH forms and read a CAR chart?
5. Can I define clean vs dirty price with a numerical example?
6. Do I remember to **write interpretation + approach first** on every question? (This is 50% of the marks.)

You have the material and the method. Show your steps, name your criterion, and the marks will follow. Good luck.